3.0.25

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Question

Let X and Y be two independent random variables. Which one of the relations between expectation(E), Variance(Var) and covariance (Cov) given below is FALSE?(ME 2007)

1. $E(XY) = E(X)E(Y)$
2. $\text{Cov}(X, Y) = 0$
3. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
4. $E(X^2Y^2) = (E(X))^2 (E(Y))^2$

Definition of Independence

X and Y are discrete random variables:

$$P(X = x, Y = y) = P(X = x)P(Y = y) \tag{1}$$

X and Y are continuous random variables:

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) \tag{2}$$

Option 1(Discrete)

$$E(XY) = \sum_x \sum_y xyP(X = x, Y = y) \tag{3}$$

$$= \sum_x \sum_y xyP(X = x)\ P(Y = y)\,(\text{independence}) \tag{4}$$

$$= \left(\sum_x xP(X = x)\right)\left(\sum_y yP(Y = y)\right) \tag{5}$$

$$= E(X)\,E(Y) \tag{6}$$

Hence TRUE.

Option 1(Continuous)

$$E(XY) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf_{X,Y}(x,y)\,dxdy \tag{7}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf_X(x)\,f_Y(y)\,dxdy \text{ (independence)} \tag{8}$$

$$= \left(\int_{-\infty}^{\infty} xf_X(x)\,dx\right)\left(\int_{-\infty}^{\infty} yf_Y(y)\,dy\right) \tag{9}$$

$$= E(X)E(Y) \tag{10}$$

Hence TRUE.

Option 2 (Discrete, Continuous)

$$\begin{aligned}
\operatorname{Cov}(X, Y) &= E(XY) - E(X)E(Y) && (11)\\
&= E(X)E(Y) - E(X)E(Y)\,(\text{from (6) and (10)}) && (12)\\
&= 0 && (13)
\end{aligned}$$

Hence TRUE.

Option 3 (Discrete, Continuous)

$$\begin{aligned}
\operatorname{Var}(X+Y) &= E\left((X+Y)^2\right) - (E(X+Y))^2 && (14)\\
&= E\left(X^2+Y^2+2XY\right) - (E(X)+E(Y))^2 && (15)\\
&= E\left(X^2\right) + E\left(Y^2\right) + 2E(XY) - \left(E(X)^2 + E(Y)^2 + 2E(X)\right. && (16)\\
&= \left(E\left(X^2\right) - E(X)^2\right) + \left(E\left(Y^2\right) - E(Y)^2\right) && (17)\\
&= \operatorname{Var}(X) + \operatorname{Var}(Y) && (18)
\end{aligned}$$

Hence TRUE.

Option 4 (Discrete)

$$E\left(X^2Y^2\right) = \sum_x \sum_y x^2y^2P\left(X = x, Y = y\right) \tag{19}$$

$$= \sum_x \sum_y x^2y^2P\left(X = x\right)P\left(Y = y\right)\text{(independence)} \tag{20}$$

$$= \left(\sum_x x^2P\left(X = x\right)\right)\left(\sum_y y^2P\left(Y = y\right)\right) \tag{21}$$

$$= E\left(X^2\right)E\left(Y^2\right) \tag{22}$$

But,

$$E\left(X^2\right) = \text{Var}\left(X\right) + \left(E\left(X\right)\right)^2 \tag{23}$$

$$E\left(Y^2\right) = \text{Var}\left(Y\right) + \left(E\left(Y\right)\right)^2 \tag{24}$$

$$\Rightarrow E\left(X^2Y^2\right) \neq \left(E\left(X\right)\right)^2\left(E\left(Y\right)\right)^2 \tag{25}$$

Option 4 (Continuous)

$$E\left(X^2Y^2\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x^2y^2 f_{X,Y}\left(x,y\right)dxdy \tag{26}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x^2y^2 f_X\left(x\right)f_Y\left(y\right)dxdy \text{ (independence)} \tag{27}$$

$$= \left(\int_{-\infty}^{\infty} x^2 f_X\left(x\right)dx\right)\left(\int_{-\infty}^{\infty} y^2 f_Y\left(y\right)dy\right) \tag{28}$$

$$= E\left(X^2\right)E\left(Y^2\right) \tag{29}$$

But,

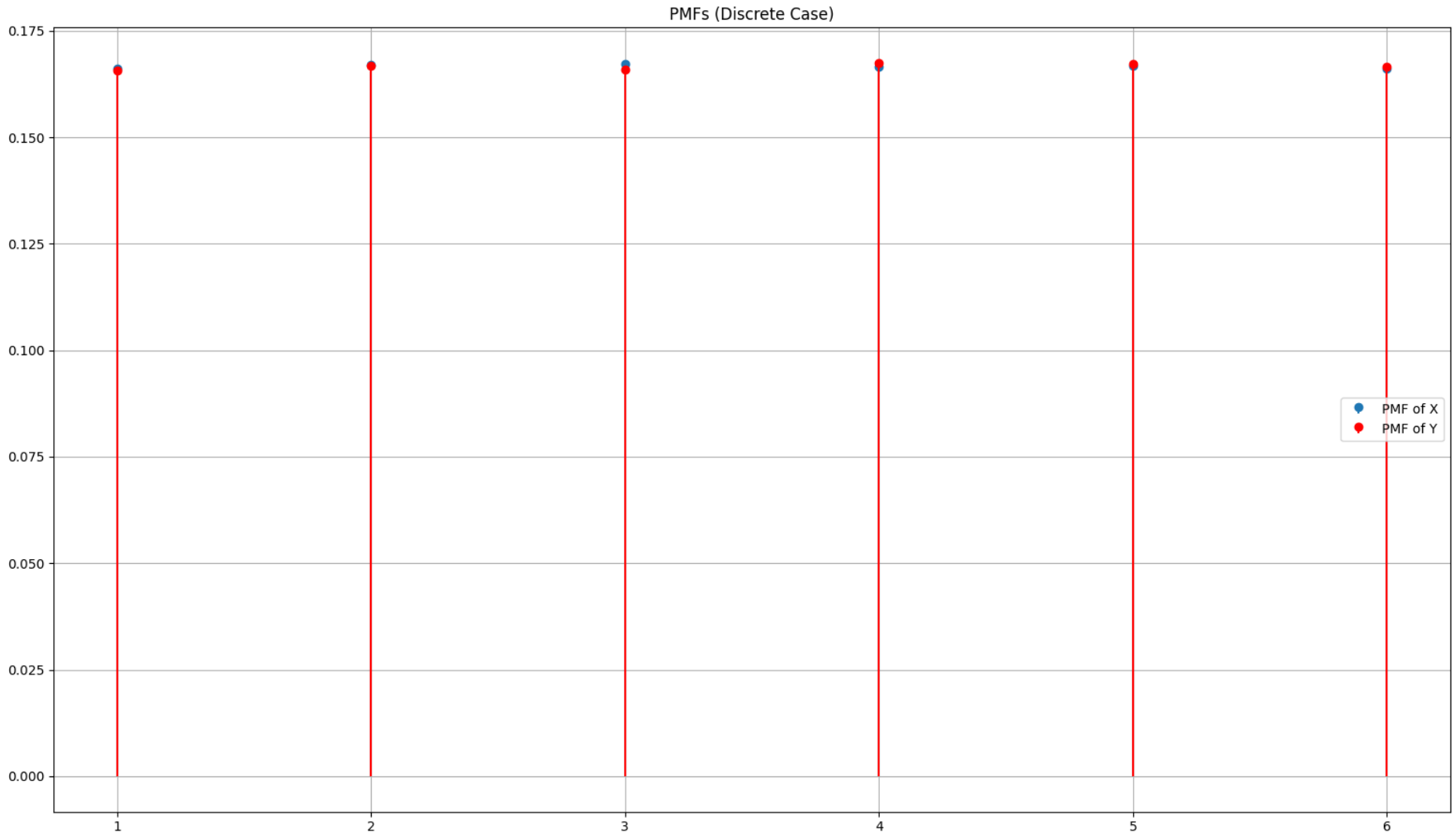
$$E\left(X^2\right) = \text{Var}\left(X\right) + \left(E\left(X\right)\right)^2 \tag{30}$$

$$E\left(Y^2\right) = \text{Var}\left(Y\right) + \left(E\left(Y\right)\right)^2 \tag{31}$$

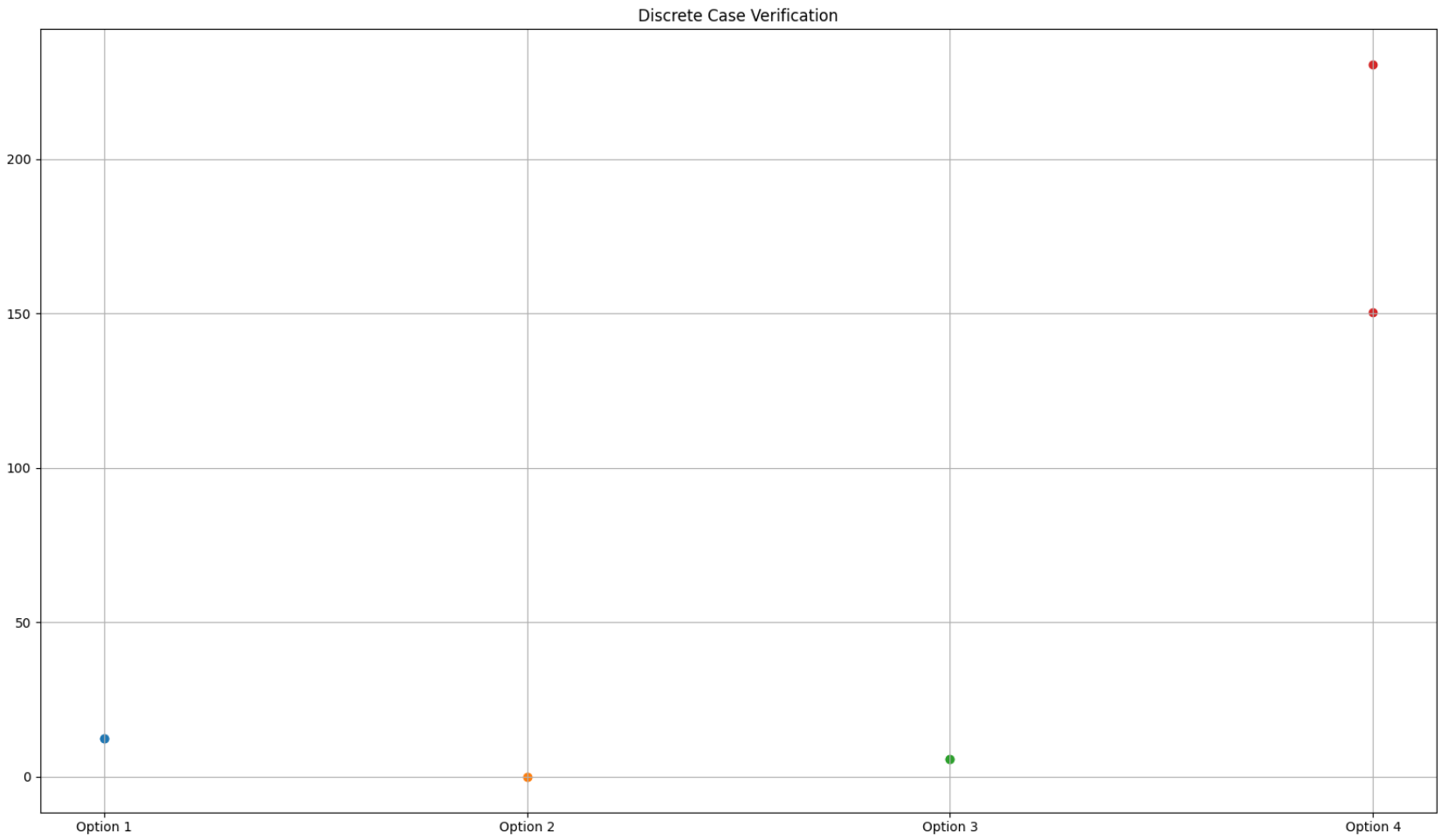
$$\Rightarrow E\left(X^2Y^2\right) \neq \left(E\left(X\right)\right)^2\left(E\left(Y\right)\right)^2 \tag{32}$$

Hence FALSE.

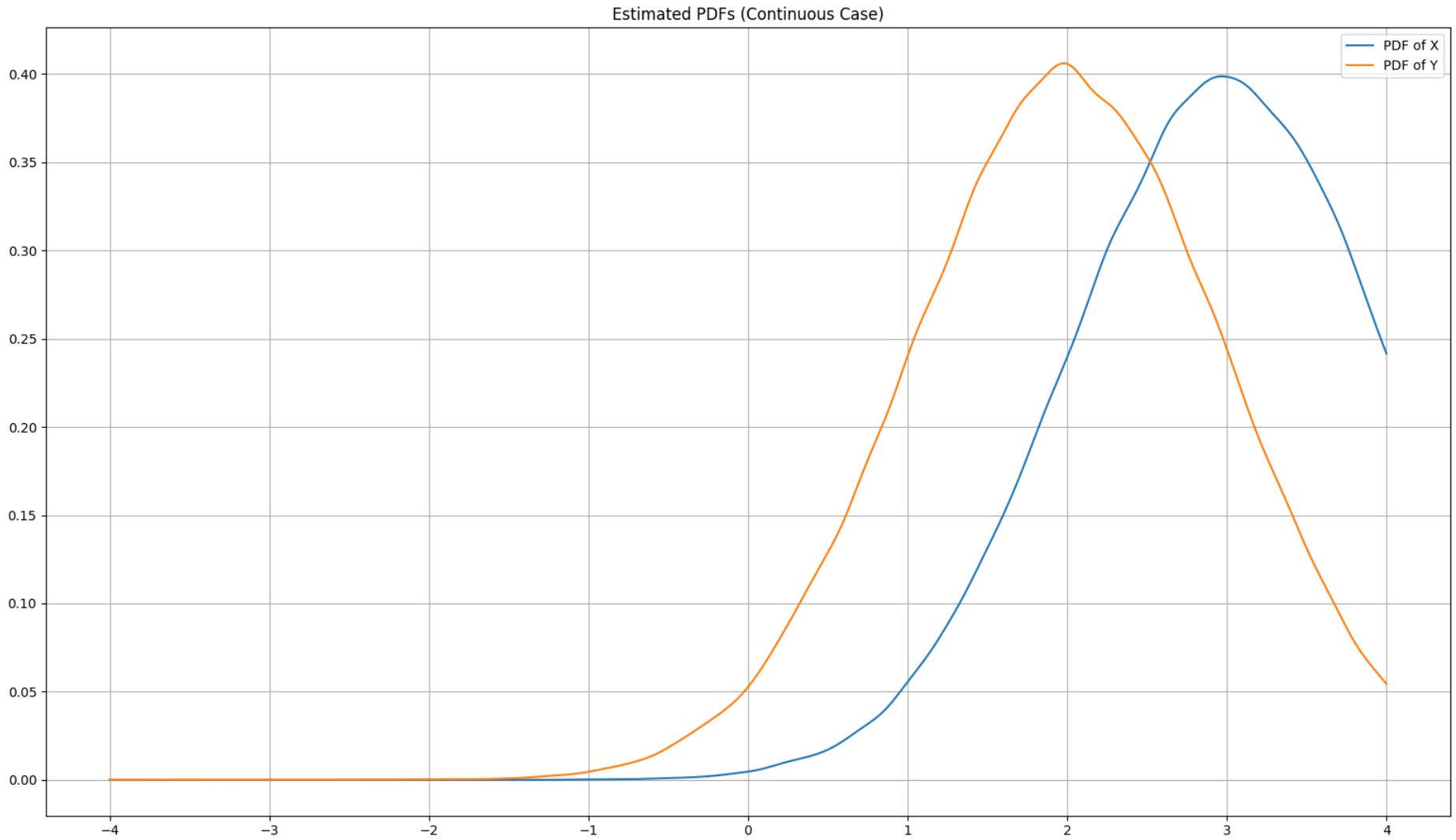
Plot-PMF


PMFs (Discrete Case)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
1
2
3
4
5
6
PMF of X
PMF of Y

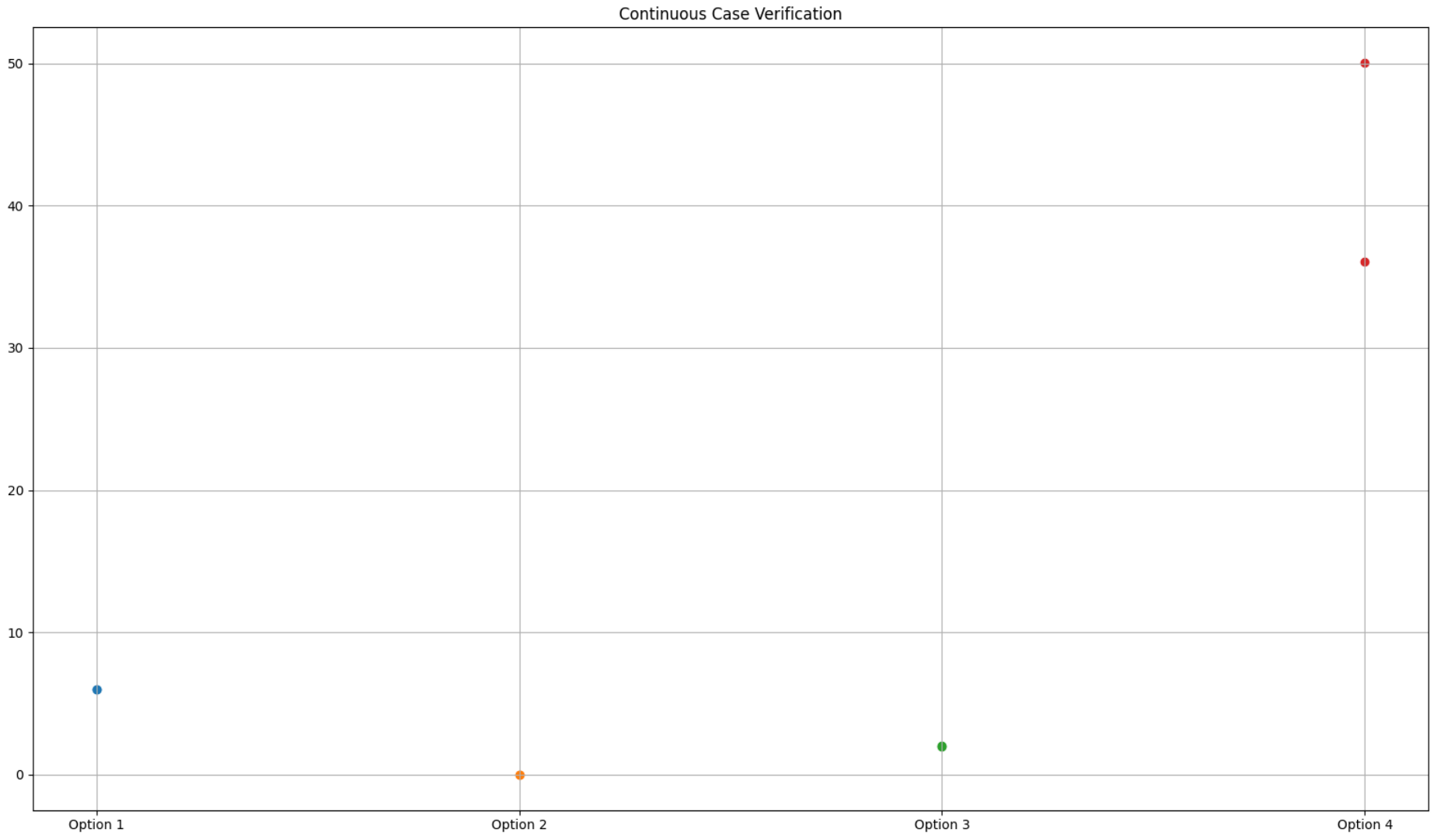
Plot-Discrete



Plot-PDF


Estimated PDFs (Continuous Case)
PDF of X
PDF of Y
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−4
−3
−2
−1
0
1
2
3
4

Continuous


Continuous Case Verification
50
40
30
20
10
0
Option 1
Option 2
Option 3
Option 4

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import gaussian_kde

# Number of samples
N = 200000

def analyze(X, Y, title):
    EX = np.mean(X)
    EY = np.mean(Y)
    EXY = np.mean(X * Y)
    cov = np.mean((X - EX) * (Y - EY))
    varX = np.var(X)
    varY = np.var(Y)
    var_sum = np.var(X + Y)
    EX2Y2 = np.mean((X**2) * (Y**2))
```

Python Code

```
# CONTINUOUS CASE (PDF)
mux = np.random.randint(1, 5, 1)
muy = np.random.randint(1, 5, 1)
Xc = np.random.normal(mux, 1, N)
Yc = np.random.normal(muy, 1, N)

results_cont = analyze(Xc, Yc, Continuous Case)

# Estimate PDFs using KDE
kde_X = gaussian_kde(Xc)
kde_Y = gaussian_kde(Yc)

x_range = np.linspace(-4, 4, 500)

plt.figure()
plt.plot(x_range, kde_X(x_range), label=PDF of X)
plt.plot(x_range, kde_Y(x_range), label=PDF of Y)
plt.title(Estimated PDFs (Continuous Case))
plt.legend()
```

Python Code

```
Xd = np.random.randint(1, 7, N)
Yd = np.random.randint(1, 7, N)

results_disc = analyze(Xd, Yd, Discrete Case)

# Compute PMFs
values_X, counts_X = np.unique(Xd, return_counts=True)
pmf_X = counts_X / N

values_Y, counts_Y = np.unique(Yd, return_counts=True)
pmf_Y = counts_Y / N

plt.figure()
plt.stem(values_X, pmf_X, label=PMF of X, basefmt= )
plt.stem(values_Y, pmf_Y, linefmt='r-', markerfmt='ro', label=PMF
    of Y, basefmt= )
plt.title(PMFs (Discrete Case))
plt.legend()
plt.grid()
```

Python Code

```
plt.figure()
for i in range(4):
    plt.scatter([i, i], cont_vals[i])
plt.xticks(range(4), labels)
plt.title(Continuous Case Verification)
plt.grid()

plt.figure()
for i in range(4):
    plt.scatter([i, i], disc_vals[i])
plt.xticks(range(4), labels)
plt.title(Discrete Case Verification)
plt.grid()

plt.show()
```